Advanced Data Analytics

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Data Analytics

- Process of exploring and analyzing data to gain insights and drive informed decision-making within organizations.
- Encompasses a wide range of techniques, methodologies, and tools aimed at leveraging data to improve business performance, enhance operational efficiency, and achieve strategic objectives.

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Different Stages in Data Analytics

- Collection of relevant data from various sources, including internal databases, external sources, and third-party data providers.
- Clean, integrate, and transform the data into a format suitable for analysis.
- Visually explore and summarize data to identify patterns, trends, and relationships.
- Apply statistical methods and models to quantify relationships between variables, test hypotheses, and make predictions.
- Forecast future outcomes or trends based on historical data and statistical algorithms

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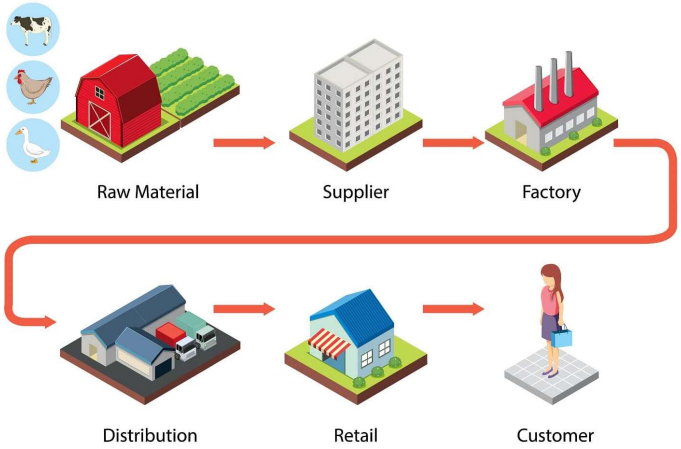
Application of Business Analytics

- Customer Segmentation
- Sales Forecasting
- Supply chain optimizatio
- Risk Analysis
- Operational Efficiency
- Predictive Maintenance



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Data Sources

- Internal Data. Eg. Customer Data, Sales data
- External Data. Eg. Social Media Platform, Market research firms, and Government agencies.
- Public Data. Eg. Census data, Weather data.
- Third-party Data: This refers to data that is purchased or acquired from third-party sources, such as data brokers or data providers.

The choice of data sources will depend on the specific objectives of the analysis and the type of insights that are needed.

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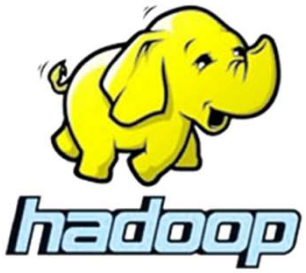
Data Sources

- Database
- Spreadsheets
- Big Data Platforms
- APIs



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Challenges

- Poor data quality, including inaccuracies, inconsistencies, and incompleteness.
- Integrating data from disparate sources with different formats and structures presents additional challenges.
- Protecting sensitive data from unauthorized access, breaches, and compliance violations.
- Challenges in recruiting and retaining qualified talent with diverse skills like programming, data analysis, statistical modelling, and domain expertise.
- Rapidly evolving landscape of analytics tools and technologies can be overwhelming for organizations to navigate and implement effectively.

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Building and Optimizing Models

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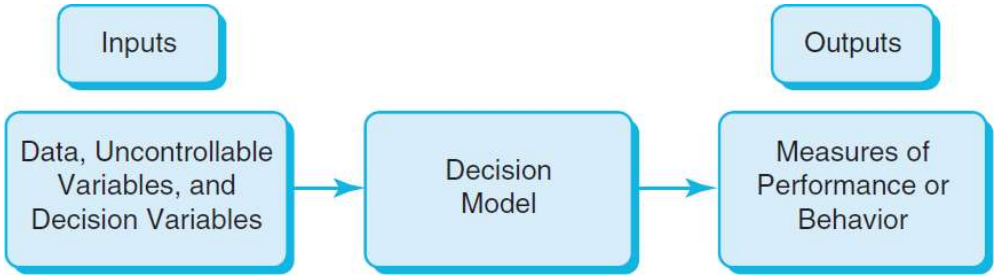
Decision Modeling



- A logical or mathematical representation of a problem or business situation that can be used to understand, analyze, or facilitate making a decision.

- Inputs:
- Uncontrollable variables, which are quantities that can change but cannot be directly controlled by the decision maker.
- Decision variables, which are controllable and can be selected at the discretion of the decision maker.

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Prescriptive Model Pricing

Predictive Analytics with Regression Techniques

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Prescriptive Model Pricing

- A firm wishes to determine the best pricing for one of its products in order to maximize profit.
- Analysts determined the following model: Sales = -2.9(price) + 3000
- Total revenue = (price)(sales)
- Fixed cost = 5000
- Total Cost = 10(Sales) + 5000

- Identify the price that maximizes profit, subject to any constraints that might exist.

Sales = -2.9 (Price) + 3000
Total revenue = (Price)(Sales)
Cost = 10(Sales) + 5000

Price	$ 100
Sales	2950
Total Cost	$ 34,500
Cost per unit	$ 12
Profit	$ 260,500

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Sales Promotion Decision

Predictive Analytics with Regression Techniques

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Sales Promotion Decision

- In the grocery industry, managers typically need to know how best to use pricing, coupons and advertising strategies to influence sales.

- Grocers often study the relationship of sales volume to these strategies by conducting controlled experiments to identify the relationship between them and sales volumes. That is, they implement different combinations of pricing, coupons, and advertising, observe the sales that result, and use analytics to develop a predictive model of sales as a function of these decision strategies.

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Sales Prediction with Multiple Regression Analysis

- Find the pricing and Ads Budget to maximize the Sales
- Use Multiple Linear Regression
- Cost price of the Product is $8
- Minimum Profit desired is $10,000

Cost Price per unit	$9.00
Minimum Profit	$4,000.00
Maximum Ads Budget	$10,000.00
Minimum Ads Budget	$1,000.00

	Coefficients
Intercept	3092.883488
Price	-142.5484493
Ads Budget	0.10225716

Results from Regression Analysis

Solution: Sales = -142 x Price + 0.10 x Ad+3091

Sales Promotion Data			
Week	Price	Ads Budget (Thousands USD)	Sales
1	$15.0	1500	1240
2	$16.4	4000	950
3	$13.0	2000	1580
4	$16.5	3000	1190
5	$12.0	1000	1430
6	$11.0	1000	1560
7	$14.3	1000	1060
8	$18.2	3465	800
9	$9.6	1000	1895
10	$15.4	4000	1475
11	$12.0	2000	1560
12	$14.0	1000	1235
13	$16.0	3000	1065
14	$15.0	1000	1025
15	$11.0	2000	1645

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Predictive Analytics with Less uncertainty

Forecasting future outcomes based on historical data and statistical techniques with minimal uncertainty.

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KERC Inc – Car Manufacturer

Optimization Model -1

Predictive Analytics with Less uncertainty

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KERC Inc – Car Manufacturer

- Has two main car models
  - KERC 501
  - KERC 601

Below table provides the Activity times (hrs) of the models

	Activity time in Hours		
Car Model	Assembly (All parts)	Frame Fitting and Painting	Quality Assurance
KERC 501	8	4	2
KERC 601	10	3	1.6

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KERC Inc – Car Manufacturer…

KERC Manufacturing Unit has a limited resources for activities as below

Manufacturing Activity	Time Available Hours per Week
Assembly (All parts)	10800
Frame Fitting and Painting	4500
Quality Assurance	2500

Car Model Profitability Table

Car Model	Profit (USD Thousands)
KERK 501	4.9
KERK 601	5.8

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Optimize KERC production

Define Objective Function (profit) and find the Optimal manufacturing quantities for car models to maximize the objective function?

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WonderZon Retailer

Optimization Model of Logistics to minimize cost

Predictive Analytics with Less uncertainty

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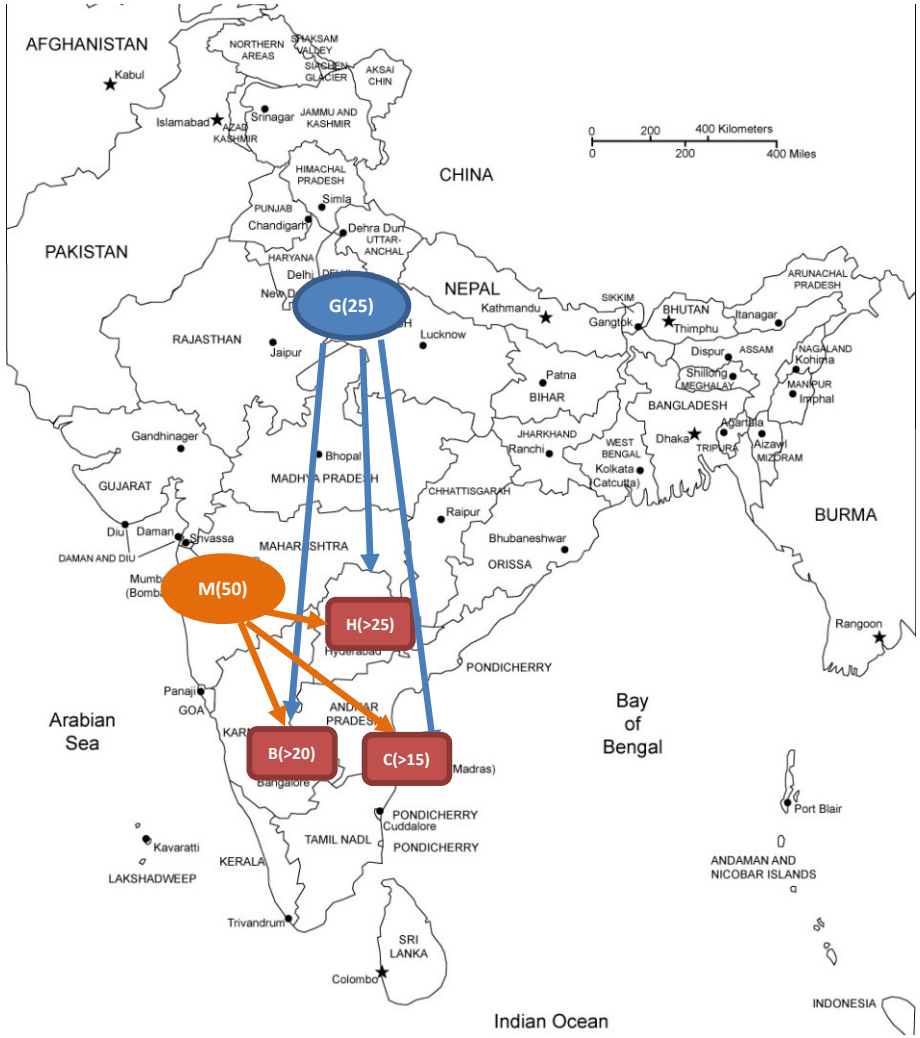
Find Objective function for cost and optimize the network for min logistics cost for WonderZon Retailer

Warehouses	Shipment Quantity
Gurgaon	25
Mumbai	50

Delivery Centers	Minimum Quantity
Bengaluru (B)	20
Hyderabad(H)	25
Chennai(C)	15

Logistics Cost in INR per Package		Delivery Centers		
	From / To	**Bengaluru (B)**	**Hyderabad (H)**	**Chennai (C)**
Warehouses	**Gurgaon (G)**	**116**	**106**	**115**
	Mumbai (M)	**110**	**95**	**112**



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Predictive Analytics

with High Uncertainty

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Internet Usage - Plan Decision

Monte Carlo Simulation

Predictive Analytics with High uncertainty

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SMART IT Inc. Internet Plan

SMART IT Inc has been using FLEXI corporate internet plan for a year and thinking about deciding to switch to FAST1000. SMART IT Inc business is seasonal and internet usage varies as per the demand.

Generally bill usage vary from around 1000 GB. Past 6 months usage in GB are 620,1050,1200,1480,870,1600

Find the best plan for SMART IT (Use Monte Carlo Simulation with Normally distribution data)

Corporate Internet Plans	
FLEXI	Flat plan charging $2 per GB
FAST1000	1000 GB included , Post 1000 GB , $1500 + $3.5 per GB

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Monte Carlo Simulation

It is a probability model used to predict the probability of various outcomes.

It is used to mitigate the possibility of unpredicted outcomes.

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Kickathlon Sport Retailer

Decision Tree & Monte Carlo Simulation

Predictive Analytics with High uncertainty

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Kickathlon – Supplier Decision Making

- Kickathlon, A French based Sports Retailer, plans to introduce a new product, TrekBareFoot – A tough Polyester based Socks, which can protect the foot while trekking barefoot.
- TrekBareFoot is scheduled to launch in next Summer beginning so should be manufactured well in advance.
- The price tag for TrekBareFoot is fixed at **$200/-**
- If the marketing efforts create a strong demand then about **15000** units can be sold, on the lower side, weak demand might end up with only **6000** units.
- There is a 50:50% chance for strong and weak demand in the market.
- TrekBareFoot being a specialized product, Kickathlon found only two suppliers one in USA and other in China. The pricing is as below.

	USA (A)	China (C)
Supplier Capacity	**6000**	**15000**
Fixed Contract Charges	**0**	**$500,000**
Unit Product cost	**$150**	**$130**
Profit per unit	**$50**	**$70**

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Decision Trees

- Decision Tree is a visual and analytical decision support tool that helps map out decisions and their possible consequences, including risks, costs and benefits.

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Decision Tree – Key Components

Decision Nodes

There are decision points where it branches to more than one decisions

Chance / Event Nodes

There are points of uncertainty, which is defined by probability / random variable

Outcome Nodes

There are the outcome variable (Objective function), which needs to be either maximized or minimized.

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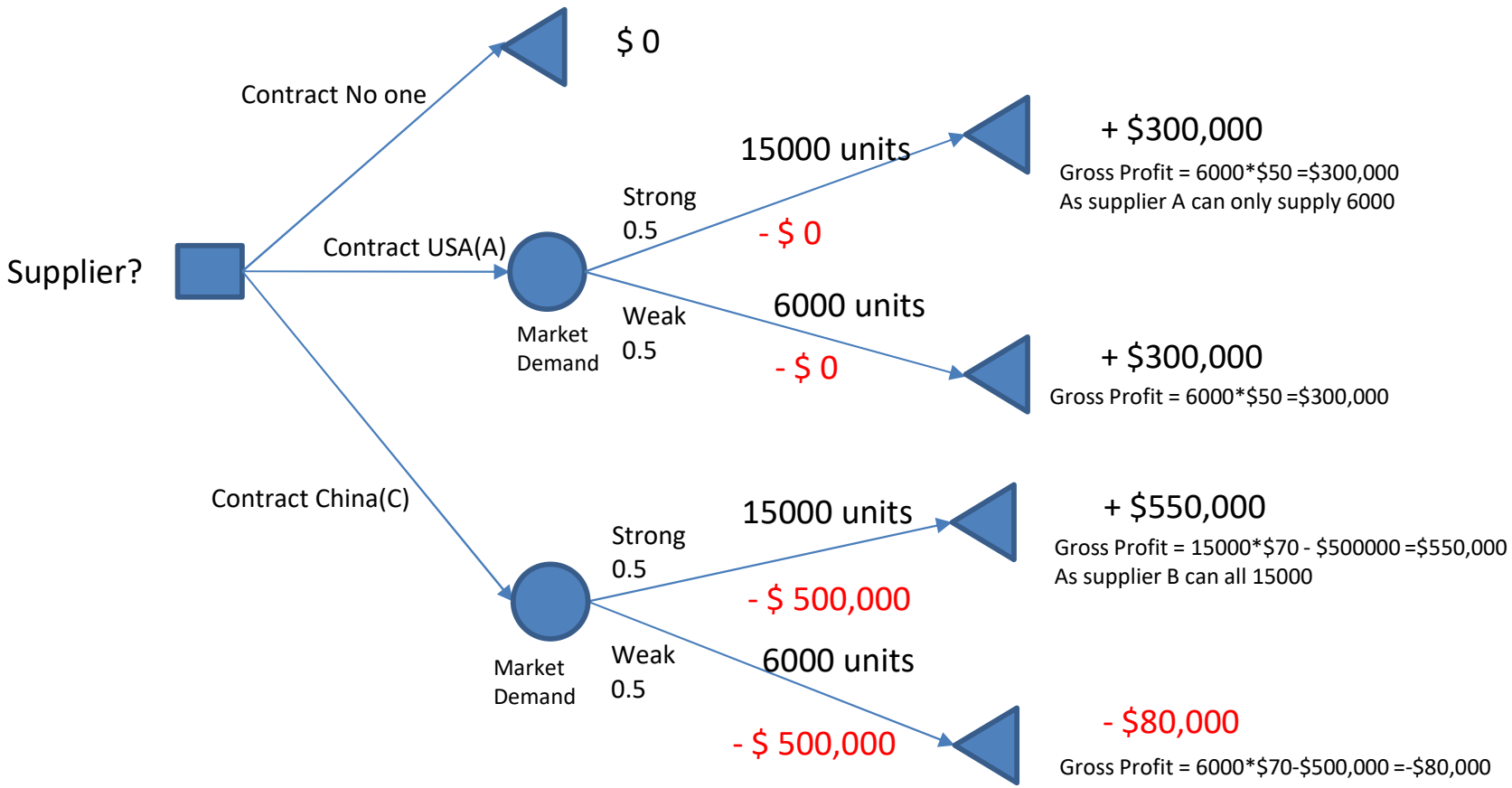
Kickathlon – Supplier Decision Tree



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Common Decision Strategies

- Maxi – Mini Strategy : Risk Averse Strategy
  - Maximize the Minimum outcome
  - Avoids bad outcomes and ignores possibility of good outcomes with risk.

- Maxi – Maxi Strategy : Risk Seeking Strategy
  - Maximize the Maximum outcome
  - Seeks good outcomes and ignore possibility of bad outcomes

- Maximize Expected Value : Risk Neutral Strategy
  - Equal consideration of Good and Bad outcomes

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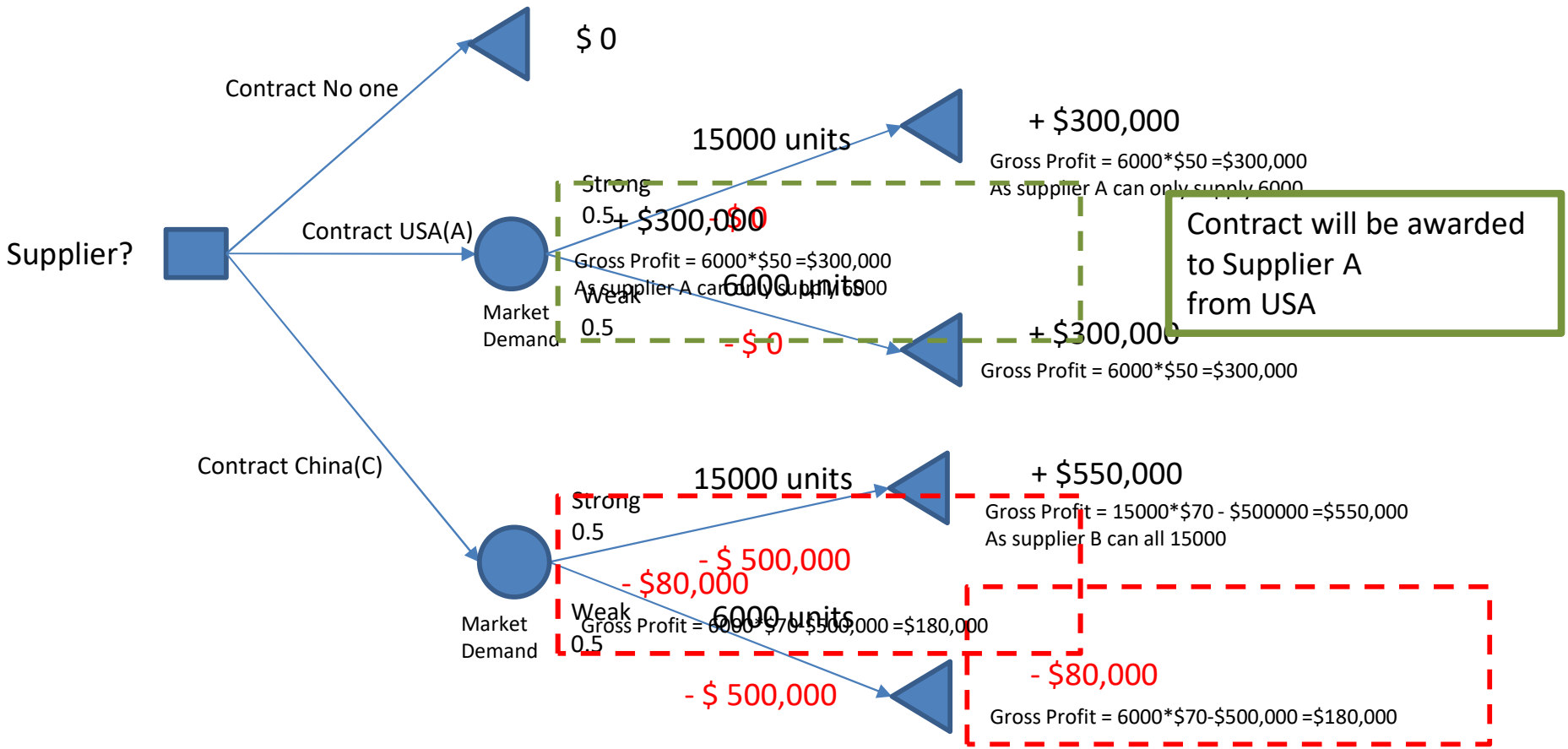
Maxi-Min Strategy



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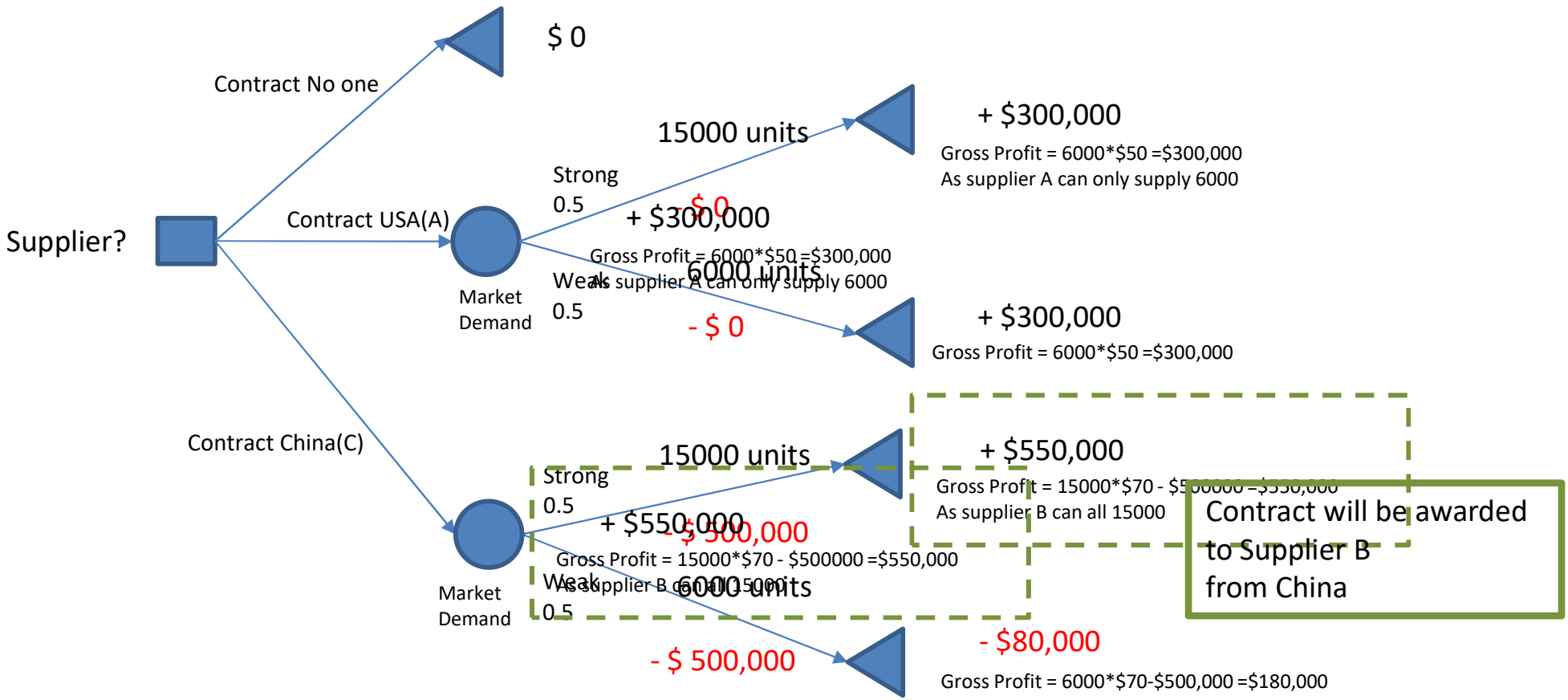
Maxi-Maxi Strategy



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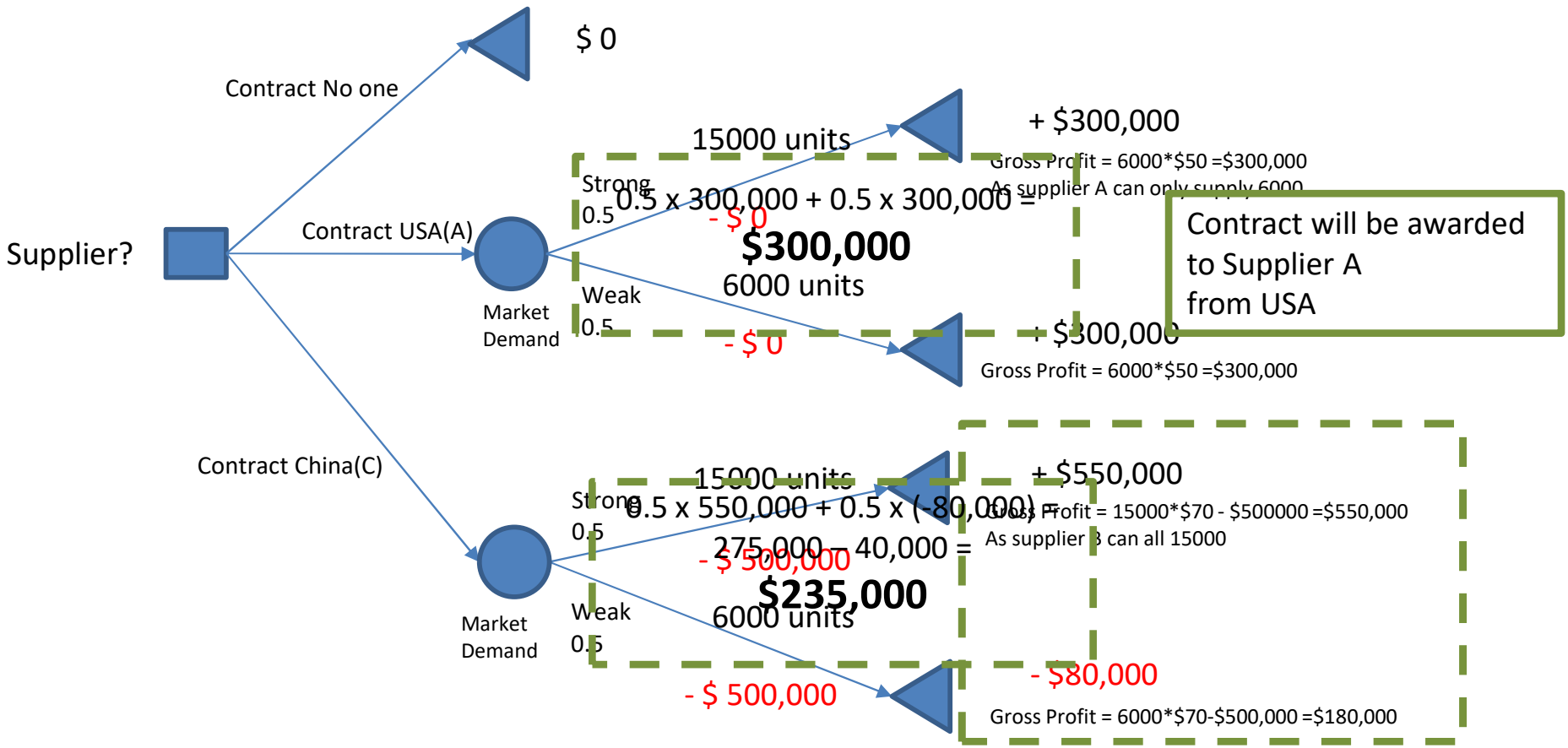
Maximize Expected Value Strategy



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What if the Demand when market in weak is not single value like 6000 units but a range like between 3000 to 8000 units?

This is the usual scenario in practice.

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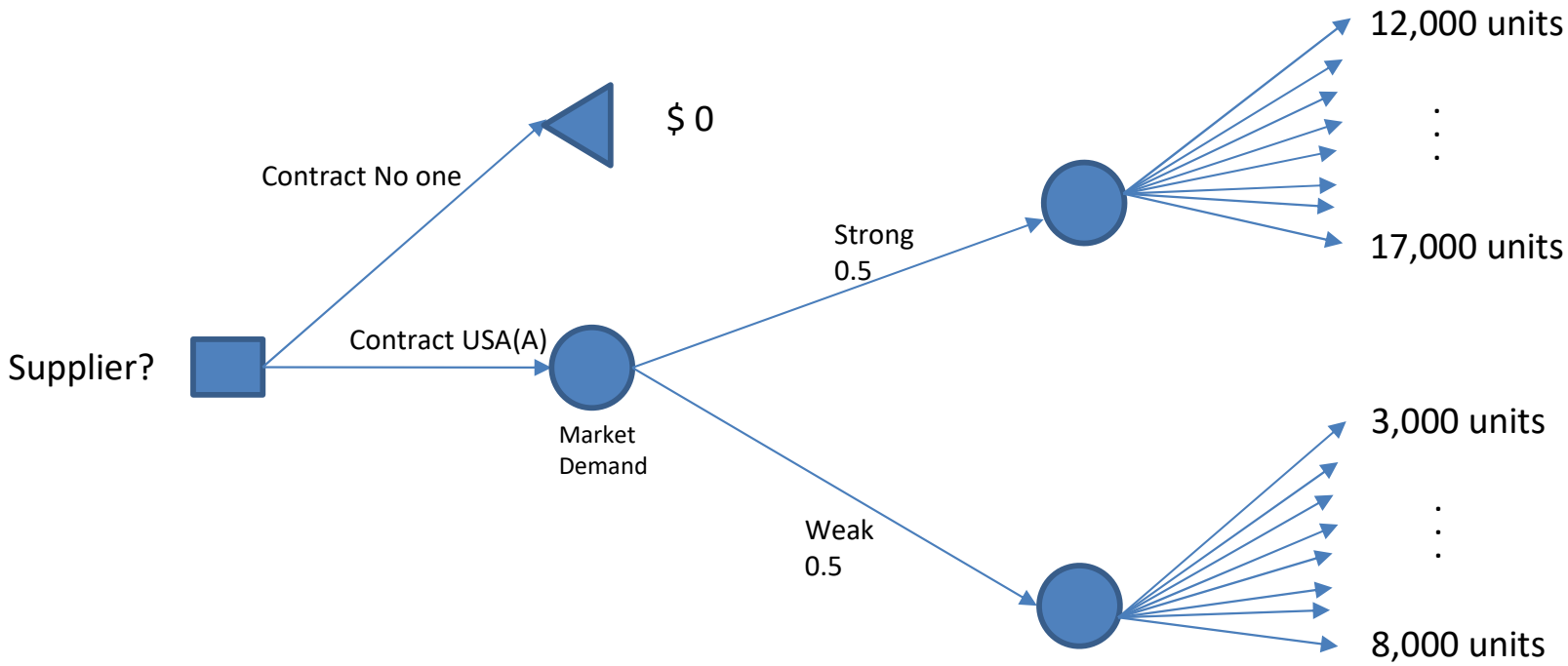

As supplier A can only supply 6000 units and unit profit is $50
Profit = $50 * min { 6000, Demand}

It is difficult to model with decision tree branching; we use Monte Carlo simulation for this kind of decision process

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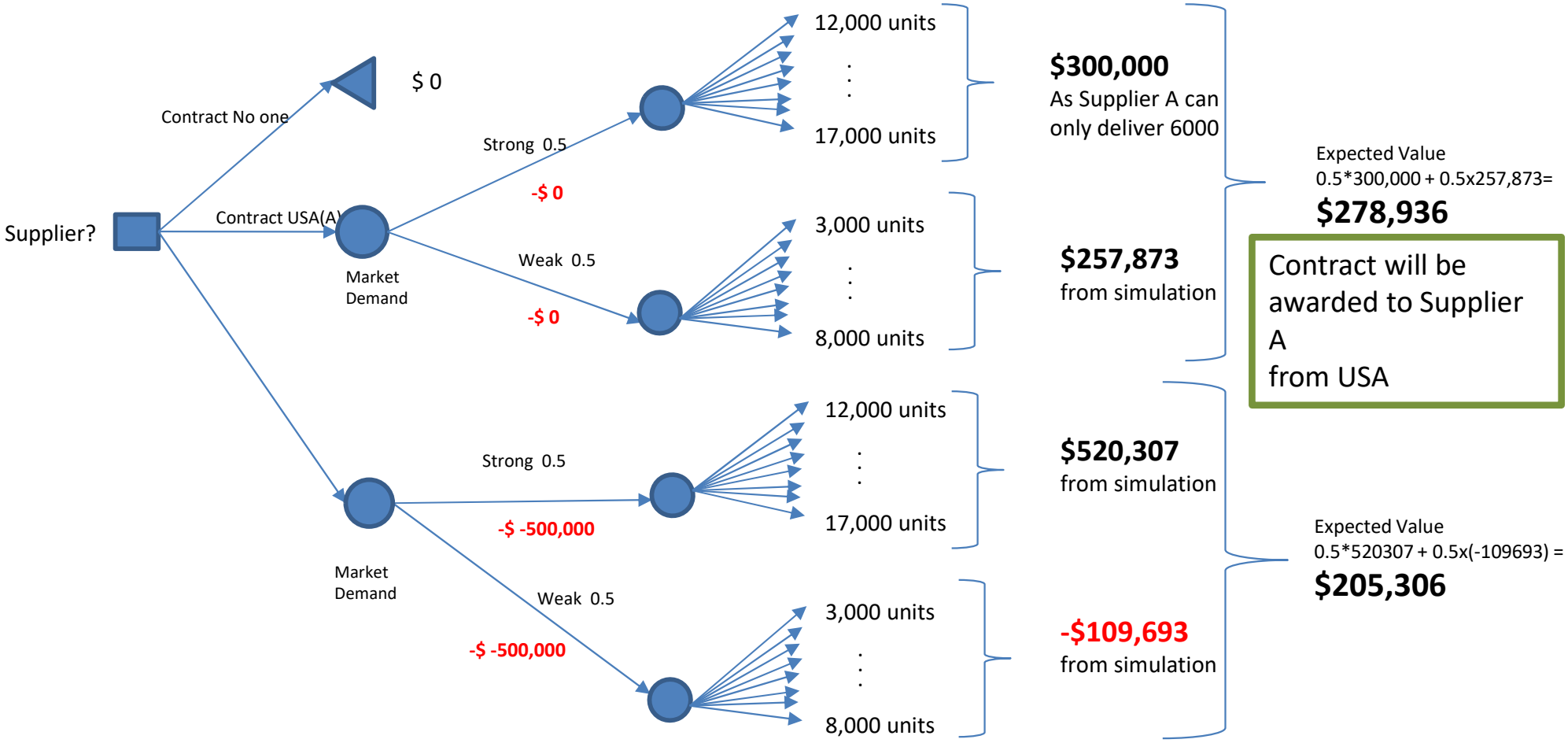

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Break Even Analysis

Decision Modeling

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Mathematical Modeling

The process of transformation business problem into a system using mathematical concepts, equations and notations is termed mathematical modeling. The Mathematical modeling is the core aspect of business analytics which enable to find optimum solutions or decisions

$$\text{Total cost} = FC + VC / \text{unit} \times Q$$

Mathematical model:

TC = Total Cost

F = Fixed cost

V = Variable unit cost

Q = Quantity produced

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Make or Buy Decision Modeling

The business can either manufacture product internally or buy from the market.

- Manufacturing has fixed cost $50,000 of setting up manufacturing facility and manufacturing cost per unit of $125.
- Buying from Market (Outsourcing) has no fixed cost but cost per unit is $175.

Decision is to choose more economic model based on volume of production.

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Break Even Decision Model

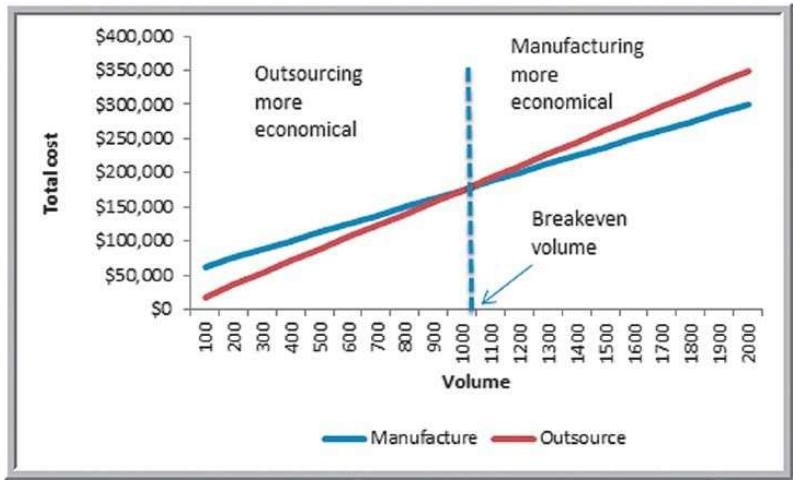
- *TC*(manufacturing) = $50,000 + $125*$Q$ *TC*(outsourcing) = $175*$Q$
- <u>Breakeven Point</u>:
  - Q where *TC*(manufacturing) = *TC*(outsourcing)

- $50,000 + $125 ×$Q$ = $175 ×$Q$
  - $50,000 = 50 ×$Q$
    - Q = 1,000

General Formula

- $F + VQ = CQ$
- $Q = F/(C - V)$

Make or Buy Decision



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What-If Analysis

What-If Analysis is a decision-making tool that helps you understand the **impact of changing input values** on the **outcome** of a model or formula.

In simple terms:

"What will happen to the result **if** I change this input?"

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Types of What-If Analysis

1. Scenario Manager

Allows you to create and compare multiple input scenarios.

Example: Best Case, Worst Case, Most Likely Case.

2. Goal Seek

Works **backward**: You know the result you want and want to find the input needed.

Example: What interest rate is needed to achieve ₹10,000 return?

3. Data Table

Shows how changing one or two inputs affects the result.

Types: One-Variable and Two-Variable Data Tables.

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Decision Trees

- Decision Trees are widely used in practice.
- Decision Tree can be built to more complexity with numerous decision points.
- Probabilities can be calculated from past data, expert judgement or Delphi method

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Why BI for Data Analyst?

- Wide range of data visualization options to communicate insights effectively to non-technical stakeholders
- Making complex data more accessible and understandable.
- To create interactive dashboards that provide real-time insights into key performance indicators (KPIs)
- To explore and interact with the data, drill down into details, and customize their views, which enhances decision-making.
- Reduces manual reporting efforts.

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Why SQL for Data Analyst?

- SQL is a fundamental skill for data analysts as it provides the means to retrieve data from relational databases.
- Also to explore, analyze, clean, transform, summarize, and merge data stored in the databases.

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End

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